Computer Programming & Engineering Analysis
CSE-125

Lab 4

Programs to understand simple conditional logic, increment decrement operators and evaluation of expressions

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Outcomes

- Things we will try to learn in this lab:
  - Know about various types of operators and their uses.
  - Learn about simple conditional logic.
  - Evaluate different types of expressions
  - Apply conditional logic to solve different problems.

Exercise 1

- Write a C program see how a prefix increment and decrement operator work.

exercise1.c

```
#include<stdio.h>
main()
{
    int x=8;
    printf("x=%d\n",x);  //8
    printf("x=%d\n",++x); //9
    printf("x=%d\n",x);  //9
    printf("x=%d\n",--x); //8
    printf("x=%d",x);  //8
}
```

Exercise 2

- Write a C program see how a postfix increment and decrement operator work.

exercise2.c

```c
#include<stdio.h>
int main()
{
    int x=8;
    printf("x=%d\n",x);   //8
    printf("x=%d\n",x++); //8
    printf("x=%d\n",x);   //9
    printf("x=%d\n",x--); //9
    printf("x=%d",x);   //8
}
```

Exercise 3

- Write a C program to find the maximum between two numbers.

exercise3.c

```
#include<stdio.h>
int main()
{
    int a, b;
    scanf("%d %d", &a, &b);
    if(a>b){
        printf("Maximum number = %d\n", a);
    }
    else {
        printf("Maximum number = %d\n", b);
    }
}
```

Exercise 4

- Write a C program to check whether a number is even or odd.

exercise4.c

```c
#include<stdio.h>
int main()
{
    int a;
    scanf("%d", &a);
    // Usage of simple if else statement
    if(a%2==0){
        printf("The number is even\n");
    }
    else{
        printf("The number is odd\n");
    }
}
```

Exercise 5

- Write a C program to check whether a character is alphabet or not.

exercise5.c

```
#include<stdio.h>
int main()
{
    char ch;
    scanf("%c", &ch);
    //Usage of else if ladder
    if(ch>='A' && ch<='Z'){
        printf("Uppercase Alphabet\n");
    } else if(ch>='a' && ch<='z'){
        printf("Lowercase Alphabet\n");
    } else{
        printf("Not a Alphabet\n");
    }
}
```

Exercise 6

- Write a C program to check whether the triangle is equilateral, isosceles or scalene triangle.

exercise6.c

```c
#include<stdio.h>
int main()
{
    int side1, side2, side3;
    printf("Enter three sides of triangle: ");
    scanf("%d%d%d", &side1, &side2, &side3);
    if(side1==side2 && side2==side3){
        printf("Equilateral triangle.");
    } else if(side1==side2 || side1==side3 || side2==side3){
        printf("Isosceles triangle.");
    } else{
        printf("Scalene triangle.");
    }
}
```

Practice Problems (Home Work)

- These problems are designed to clear your basic understanding on operators, expressions and basic conditional logic.

Practice Problem 1

- Write a C program to check whether a number is negative, positive or zero.

Practice Problem 2

- Write a C program to check whether a year is leap year or not.

Practice Problem 3

- Write a C program to input any character and check whether it is alphabet, digit or special character.

Practice Problem 4

- Write a C program to input any alphabet and check whether it is vowel or consonant.

Practice Problem 5

- Write a C program to input all sides of a triangle and check whether triangle is valid or not.

Practice Problem 6

- Write a C program to input marks of five subjects Physics, Chemistry, Biology, Mathematics and Computer. Calculate percentage and grade according to following: (Use else if Ladder)
- Percentage >= 90% : Grade A
- Percentage >= 80% : Grade B
- Percentage >= 70% : Grade C
- Percentage >= 60% : Grade D
- Percentage >= 40% : Grade E
- Percentage < 40% : Grade F

Practice Problem 7

- Find the output of the following program.

```
#include<stdio.h>
main()
{
    int x =100;
    printf("%d\n", 10 + x++);
    printf("%d\n", 10 + ++x);
}
```

Practice Problem 8

- Find the output of the following program.

```
#include<stdio.h>
main()
{
    int x =5, y=10, z=10;
    x = y==z;
    printf("%d", x);
}
```

Special Notes

- Please read your text book to clear your understanding.
- Follow reference Book and do the exercise and examples of the book.

Thank You

Questions & Discussion Welcome

Feel free to reach out after class or during office hours.

C Programming